

Optimal Cardiovascular Pulse Sequences and Statistical Conformal Coils using Ramanujan Operator Signatures

Cartik Sharma, Neuromorph System

Abstract: Cardiovascular magnetic resonance imaging is inherently bounded by dynamic tissue motion and non-Euclidean anatomic geometry. In this report, we formalize the generation of optimal statistical conformal coil arrays governed by Ramanujan tau properties. We couple these geometries with newly synthesized pulse sequences defined via finite difference methods. These sequence and coil combinations maximize signal-to-noise ratio over moving cardiac topologies, establishing a rigorous finite mathematical standard for cardiovascular structural evaluation.

1. Statistical Conformal Coil Geometry

The conformal mapping of the receiver array over the bi-ventricular surface is designed globally to mitigate B1 field inhomogeneities. The node positions are defined statistically using a Ramanujan operator approach, wherein the density of nodes correlates with prime-gap distributions:

$$\tau(n) = O(n^{11/2}), \text{ where } \tau \text{ denotes the Ramanujan tau function}$$

The finite mapping function Φ discretizes the non-Euclidean manifold such that the distance metric $d(s)$ between adjacent resonating elements remains strictly conformal under dynamic ventricular contraction.

2. Finite Operator Approximation for Coil Signatures

We attribute the signal reception efficiency to the distribution signature of the constructed coils. The induced voltage integral is solved via finite summation over the M independent nodes:

$$V_{sig} \approx \sum_{m=1}^M \int B_{1,m} \cdot \partial M_t / \partial t dV$$

This summation acts as a discrete filter, ensuring noise signals are orthogonally rejected based on the Ramanujan-weighted node locations.

3. Synthesized Pulse Sequences for the Ramanujan Array

The optimal radiofrequency pulses designed for this specific conformal array adopt a statistical variance model. We apply finite state gradient ascent to determine the amplitude sequence:

$$B_1(t_k) = \mu + \sigma \cdot \tau(k) \cdot \Delta t$$

The sequence intrinsically compensates for phase dispersion arising from the heart's moving boundaries, stabilizing the discrete magnetization matrix $\nabla^2 M$ at each finite coordinate step.